

CONTACT INFORMATION	84 Wood Lane London W12 0BZ, UK	<code>firstinitial.lastname18@imperial.ac.uk</code> <code>https://sloomba.github.io</code>
RESEARCH INTERESTS	Causal inference, statistical network modelling, probability on graphs, partially observed data, Bayesian inference, statistical machine learning, computational social science, systems biology, network spillovers	
EMPLOYMENT	Imperial College London, UK Chapman–Schmidt AI in Science Fellow Oct 2025 – PI: Prof. Mauricio Barahona, Imperial Mathematics Massachusetts Institute of Technology, US Schmidt Science Fellow at Institute for Data, Systems, and Society Oct 2023 – Sep 2025 PIs: Prof. Dean Eckles, MIT Sloan; Prof. Elchanan Mossel, MIT Mathematics Imperial College London, UK EPSRC Doctoral Prize Fellow in Mathematics Oct 2022 – Sep 2023 Graduate Teaching Assistant in Mathematics Oct 2018 – Dec 2019 Harvard University, US Predoctoral Fellow in Systems Biology at the Wyss Institute Aug 2016 – Jul 2018 PI: Prof. James J Collins, MIT Biological Engineering Xerox Research Center India, IN Research Intern in Multimedia Analytics May – Jul 2015	
EDUCATION	Imperial College London, UK PhD in Mathematics Oct 2018 – Sep 2022 PI: Prof Nick S Jones Thesis: Sparse and partially observed large-scale networks: Analytic statistics, behaviour, and structural inference Indian Institute of Technology Delhi, IN BTech in Computer Science and Engineering 2012 – 2016 PI: Prof Sumeet Agarwal Thesis: Causal computational models for gene regulatory networks	
HONORS	Chapman–Schmidt AI in Science Fellowship 2025 Yael Naim Dowker Prize <i>for the best Mathematics PhD thesis</i> 2024 Schmidt Science Fellowship <i>for the world’s best emerging scientists to pursue a postdoctoral fellowship</i> 2023 EPSRC Doctoral Prize Fellowship <i>for the very best newly qualified PhDs receiving EPSRC doctoral funding</i> 2022 Doris Chen Mobility Award <i>for PhD students with exceptional potential to take their research abroad</i> 2022 Imperial–TUM Global Fellows Programme 2019 Cargill Global Scholars Award <i>for academic excellence & leadership</i> 2014 IIT Delhi Institute Medal <i>for obtaining highest overall GPA</i> 2013 IIT Delhi Merit Semester Scholarship 2012, 2013	
PUBLICATIONS	STATISTICAL THEORY & METHODS • Loomba, S. & Eckles, D. (2025). Policy relevance of causal quantities in networks. <i>arXiv preprint under R&R at Harvard Data Science Review</i>, arxiv:2507.14391. We show how causal estimands under network interference can be seen as two kinds of averaging: one can be interpreted as unit-level causal effects but the other is more policy-relevant. • Loomba, S. & Jones, N. S. (2025). Geodesic length distribution in sparse network ensembles. <i>arXiv preprint under R&R at Physical Review X</i>, arxiv:2111.02330v2. We derive the analytic distribution of distances in super- and sub-critical regimes in a large family of conditionally independent edge models that includes stochastic block models and graphs.	

- Rubin, J., **Loomba, S.**, & Jones, N. S. (2025). Limits of message passing for node classification: How class-bottlenecks restrict signal-to-noise ratio. *arXiv preprint under submission at PNAS*, arxiv:2508.17822.

We give a statistical perspective on MPNN performance for node classification by decomposing it into signal and noise sensitivity and showing that the former relates to higher-order homophily.

APPLIED STATISTICS AND MODELLING

Social sciences

- **Loomba, S.**^{*}, de Figueiredo, A.[†], Piatek, S. J., de Graaf, K., & Larson, H. J. (2021). Measuring the impact of COVID-19 vaccine misinformation on vaccination intent in the UK and USA. *Nature Human Behaviour*, 5(3), 337-348, doi:10.1038/s41562-021-01056-1.

Through a randomized controlled study, we show that even brief exposure to COVID-19 vaccine misinformation can reduce the intention to definitely vaccinate by as much as 6.2%.

- **Loomba, S.**[†], Maertens, R., Roozenbeek, J., Götz, F., van der Linden, S., & de Figueiredo, A. (2023). Ability to detect fake news predicts sub-national variation in COVID-19 vaccine uptake across the UK. *medRxiv preprint under R&R at Nature Human Behaviour*, doi:10.1101/2023.05.10.23289764.

Using a psychometrically-validated test of misinformation susceptibility, we show that belief in fake news is credibly associated with lower actual COVID-19 vaccine uptake at the national level.

- Götz, F., Maertens, R., **Loomba, S.**, & van der Linden, S. (2023). Let the Algorithm Speak: How to Use Neural Networks for Automatic Item Generation in Psychological Scale Development. *Psychological Methods*, doi:10.1037/met0000540

We leverage recent advances in transformer-based neural language models that generate human-like text, like GPT, for automatic item generation for psychological scales.

Life sciences

- Novak, R., Lin, T., Kaushal, S., Sperry, M., Vigneault, F., Gardner, E., **Loomba, S.**, Shcherbina, K., Keshari, V., Dinis, A., Vasan, A., Chandrasekhar, V., Takeda, T., Turner, J., Levin, M., & Ingber, D. (2025). AI-enabled drug prediction and gene network analysis reveal therapeutic use of vorinostat for Rett Syndrome in preclinical models. *Communications Medicine*, 5(259)doi:10.1038/s43856-025-00975-8.

We combine network-based computational modelling with *in vivo* screening in a CRISPR-edited *Xenopus laevis* tadpole model of Rett syndrome to carry out target-agnostic drug discovery.

- Kite, K. A., **Loomba, S.**, Elliott, T. J., Yongblat, F., Lightbown, S. L., Doyle, T. J., Gates, L., Alber, D., Downey, G. A., McCurdy, M. T., Hill, J. A., Super, M., Ingber, D. E., Klein, N., & Cloutman-Green, E. (2022). FcMBL magnetic bead-based MALDI-TOF MS rapidly identifies paediatric blood stream infections from positive blood cultures. *PLOS ONE*, doi:10.1371/journal.pone.0276777

We show that broad-range pathogen binding properties of an engineered FcMBL enable more accurate and rapid identification of patient positive cultures than the state-of-the-art.

Miscellaneous

- Gandhi, A., Biswas, A., Shrivastava, K., Kumar, R., **Loomba, S.**, & Deshmukh, O. (2016, March). Easy Navigation through Instructional Videos using Automatically Generated Table of Content. In *Companion Publication of the 21st International Conference on Intelligent User Interfaces* (pp. 92-96). ACM, doi:10.1145/2876456.2879472.

We develop a computational pipeline to automatically generate a table of content for video lectures, using saliency scores of visual and spoken words for topic-based segmentation.

PUBLICATIONS IN PREPARATION

STATISTICAL THEORY & METHODS

- **Loomba, S.** & Eckles, D. (In prep). Off-policy Causal Estimation in Networks. *In prep for Annals of Statistics*.

Under network interference, causal estimands are functions of the treatment policy. We provide consistent statistical estimators for off-policy causal estimation under weak notions of sparsity.

- **Loomba, S.** & Jones, N. S. (In prep). Geodesic-Based Percolation and Centrality Statistics in Sparse Network Ensembles. *arXiv preprint in prep for Physical Review X*, arxiv:2111.02330v1.

We leverage the analytic distribution of shortest path lengths to yield analytical results on the percolation threshold and path-based centralities in sparse conditionally independent edge models.

^{*}Co-first author

[†]Corresponding author

APPLIED STATISTICS AND MODELLING

- Araujo, A. M., Sarker, A., Eckles, D., **Loomba, S.**[†] (In prep) Disentangling dyadic and triadic social network effects. *In prep for Nature Human Behaviour*.

Through a large-scale survey sample we elicit data on group social interactions to understand the socio-spatial mechanisms that drive them, and their differences from pairwise interactions.

- Piatek, S. J., **Loomba, S.**^{*}, Wiegand, M., Dek, I., Pertwee, E., Paterson, P., Jarret, C., Larson, H.J. (In prep) Misinformation impacts beliefs about democratic values, climate change, and COVID-19 vaccinations: Evidence from Nigeria, Germany, and Poland. *In prep for Nature Human Behaviour*.

Through a large-scale randomized trial we find that misinformation negatively impacts belief in democratic values with individual heterogeneity across political affiliations across countries.

- Happenhofer, J., **Loomba, S.**, Hoffman, T., Agarwal, S., Jones, N. S. (In prep). Social and geographic segregation jointly limit opportunities for racial and income mixing in the US. *In prep for Nature Human Behaviour*.

We generalise social segregation measures via f -divergences of conditional distributions over a socio-physical space and infer them from geo-aggregated Facebook friendship data in the US.

PATENTS

- Cartwright, M. J., Super, M., Ingber, D. E., Grant, J., Scott, J., Duffy, S. C., & **Loomba, S.** (2021). Methods for detection of microbes and microbe components. *International Publication No. WO 2021/201949 A2*.

Technology licensed to *BOA Biomedical*, with the aim of rapid detection of active infection from a blood sample within 1 hour, which is crucial for treating sepsis.

- Barhate, S. S., **Loomba, S.**, Gandhi, A., Biswas, A., Negi, S., & Deshmukh, O. D. (2019). Method and system for generation of a table of content by processing multimedia content. *U.S. Patent No. US 10,296,533 B2*.

Technology licensed to *VideoKen* for automated video content navigation.

- Gandhi, A., Biswas, A., Deshmukh, O. D., & **Loomba, S.** (2018). Method and system for content processing to determine pre-requisite subject matters in multimedia content. *U.S. Patent Publication No. US 2018/0060984 A1*.

Technology licensed to *VideoKen* to create course structures for effective learning.

GRANTS

Mechanistically informed reaction embeddings for biocatalysis

Schmidt Science Fellows, Senior Fellows Catalyst Grant, \$10K

Oct 2025

co-PI: Dr Ariane Mora, AITHYRA Research Institute for Biomedical Artificial Intelligence

Enhancing enzymatic reaction embeddings derived from large language models by explicitly encoding mechanistic understanding through a non-reductive design approach.

Develop a unified, compositional representation of enzymatic catalysis that captures the interplay of reaction components, treating them as more than just the sum of their parts.

Synergistic Discovery and Design (SD2)

DARPA/I20, HR001117S0003, \$2M

Sep 2017

PI: Prof James J Collins, MIT Biological Engineering

Co-authored research proposal for data-driven discovery and design of biological circuits.

Contributed about half of each of the writing, conceptual and methodological content.

Designed core algorithmic pipeline to generate testable circuit designs: going from high-throughput omics data to inferred network structures, to functional network motifs, down to operable biological circuits in traditional and novel host organisms.

TEACHING

Developed and delivered a lecture for

Research Design: Decision-Theoretic Perspectives (PhD; MIT Sloan Marketing)

Nov 2025

Seminar on *Introduction to applied Bayesian inference* (Imperial-X)

Apr 2023

Developed and delivered two lectures for

Methods for Data Science (Years 3 & 4, MSc; Imperial Maths)

Jan – Mar 2023

GRADUATE TEACHING (TUTORIALS)

Methods for Data Science (Years 3 & 4, MSc; Imperial Maths)

Oct – Dec 2019

Mathematical Computation in Python (Year 1; Imperial Maths)

Jan – Mar 2019

Probability and Statistics II (Year 2; Imperial Maths)

Oct – Dec 2018

SUPERVISION	Alejandro Araujo	Year 2, Bachelors in Computer Science, Dartmouth	2025 –
	Jonathan Rubin	PhD in Computing, Imperial College London	2023 – 2025
		Year 4, MSci in Mathematics, Imperial College London	2022 – 2023
	Amelia French	Year 4, MSci in Mathematics, Imperial College London	2021 – 2022
	Kushal Kejriwal	Year 2, Dual degree in Electrical Engineering, IIT Bombay	2021
	Siddharth Chandak	Year 2, BTech in Electrical Engineering, IIT Bombay	2019
	Katie Collins	Year 1, Bachelors in Brain and Cognitive Sciences, MIT	2017

PRESENTATIONS

TALKS

- **Loomba, S.** & Eckles, D. (2026, November). Policy relevance of causal quantities in networks. Invited presentation at *ASA Statistical Learning and Data Science Conference*.
- **Loomba, S.** & Eckles, D. (2025, November). Policy relevance of causal quantities in networks. Accepted presentation at *Conference on Digital Experimentation*.
- **Loomba, S.** & Eckles, D. (2025, August). Off-policy causal estimation in social network studies. Invited presentation at *Joint Statistical Meeting*.
- **Loomba, S.** & Eckles, D. (2025, June). Off-policy causal estimation under arbitrary network interference. Invited presentation at *INFORMS Applied Probability Society Conference*.
- **Loomba, S.** & Eckles, D. (2025, May). Off-policy causal estimation in social network studies. Invited presentation at *American Causal Inference Conference*.
- **Loomba, S.** & Eckles, D. (2025, May). Off-policy causal estimation in social networks. Invited presentation at *MIT IDE Lunch Seminar*.
- **Loomba, S.** & Eckles, D. (2025, April). Off-policy causal estimation in networks. Invited presentation at *Network Science Institute at Northeastern University*.
- **Loomba, S.** & Eckles, D. (2024, October). Off-policy causal estimation under arbitrary network interference. Invited presentation at *Conference on Digital Experimentation*.
- **Loomba, S.** (2024, March). Social covariation and spillovers using geo-aggregated network data. Invited presentation at *MIT Sloan Marketing Seminar*.
- **Loomba, S.[†]**, Happenhofer, J.[†], Hoffmann, T., Agarwal, S., Jones, N. S. (2023, July). Inferring socio-physical networks and social Gini indices from aggregated data. Accepted presentation at *NetSci Satellite Symposium on Statistical Inference for Network Models*.
- **Loomba, S.** (2023, June). Fake news and global health. Invited panel discussion at *The London University Diplomatic Summit*.
- **Loomba, S.[†]**, Happenhofer, J.[†], Hoffmann, T., Agarwal, S., Jones, N. S. (2022, July). Global mapping of a “social Gini” index using socio-physical connectivity kernels inferred from large-scale social network data. Accepted presentation at *8th International Conference on Computational Social Science*.
- **Loomba, S.** (2021, November). Misinformation and the pandemic: a tale of two contagions. Invited presentation at *Students for Global Health Conference 2021*.
- **Loomba, S.** & Jones, N. S. (2021, July). Geodesic statistics for random network families. Accepted presentation at *Networks 2021 (Joint Sunbelt and NetSci Conference)*.
- **Loomba, S.** & Jones, N. S. (2020, July). Scalable statistics for social networks of entire societies. Accepted presentation at *Sunbelt XL Conference*.
- **Loomba, S.** & Jones, N. S. (2020, March). Stochastic block models as a social connectivity kernel. Invited presentation at *CMPH Seminar, Imperial College London*.
- **Loomba, S.** (2018, August). Mathematical Representations For Biological Systems. Invited presentation at *DAIR Seminar, Indian Institute of Technology Delhi*.

POSTERS

- **Loomba, S.** & Eckles, D. (2026, January). Policy relevance of causal quantities in networks. Accepted presentation at *Causal Inference Workshop at Isaac Newton Institute for Mathematical Sciences*.
- **Loomba, S.** & Eckles, D. (2025, June). Off-policy causal estimation under network interference. Accepted presentation at *ACM SIGMETRICS Causal Inference Workshop*.
- **Loomba, S.** & Eckles, D. (2025, April). Off-policy causal estimation under social network interference. Accepted presentation at *Network Science and Economics Conference*.
- **Loomba, S.** (2023, July). Causal inference in sparse and partially observed large-scale social networks. Accepted presentation at *9th International Conference on Computational Social Science*.
- **Loomba, S.** (2023, July). Causal inference in sparse partially observed large-scale networks. Accepted presentation at *International School and Conference on Network Science*.

- **Loomba, S.**, Hoffmann, T., & Jones, N. S. (2019, July). Social Access Statistic: Linking Social Connectivity to Health Outcomes. Accepted presentation at *5th International Conference on Computational Social Science*.
- **Loomba, S.** & Garrod, M. (2019, July). How Far Would You Go? Comparing Urban Access in 10 Global Cities. Accepted presentation at *NetMob 2019*.

IMPACT

Generated viable drug candidate(s) for treatment of a monogenic disease

Created *NeMoCAD*, a Network Model for Causally Aware Discovery, that learns from gene-regulation and perturbative drug-gene interaction data to query for desirable therapeutic states; discovered a new therapeutic for Rett Syndrome which was validated in mouse models.

Software licensed to Unravel Biosciences

Impacted public health policy on health misinformation

Research on the impact of COVID-19 vaccine misinformation on vaccine uptake has seen considerable impact on pandemic research (2200+ citations), and on policymaking: the US Surgeon General's health misinformation advisory cites our work, and I delivered a video message for the UN on the impact of misinformation on our ability to end the pandemic.

POPULAR PRESS

Times Higher Ed Understanding complexity with computation and interdisciplinary science

UN Verified Is online misinformation threatening our ability to end the pandemic?

Reuters Misinformation could prompt people to turn against COVID-19 vaccines: study

BBC Covid vaccine: Rumours thrive amid trickle of pandemic facts

The Times Herd immunity put at risk by a few lies about Covid vaccine

The Quint Exposure to Misinformation Reduces COVID Vaccine Acceptance: Study

Hindustan Times Misinformation may hit vaccination drive: Study

SERVICES

CONFERENCE ORGANIZER

CausNetS: A Causal Network Science Satellite at NetSci 2026

PROGRAM COMMITTEE MEMBER

Conference on Digital Experimentation 2024

SFE-TSDM at IEEE International Conference on Data Mining 2021

REVIEWING

Journal of Marketing Research 2025 –

Physical Review Research, Journal of the Royal Statistical Society: Series B 2024 –

Epidemics, Journal of Physics: Complexity 2023 –

Science, Healthcare in Low-resource Settings, Mathematical Modelling of Natural Phenomena 2022 –

PNAS, New Journal of Physics, Vaccine, Social Network Analysis and Mining 2021 –

Physical Review E, Scientific Reports 2019 –